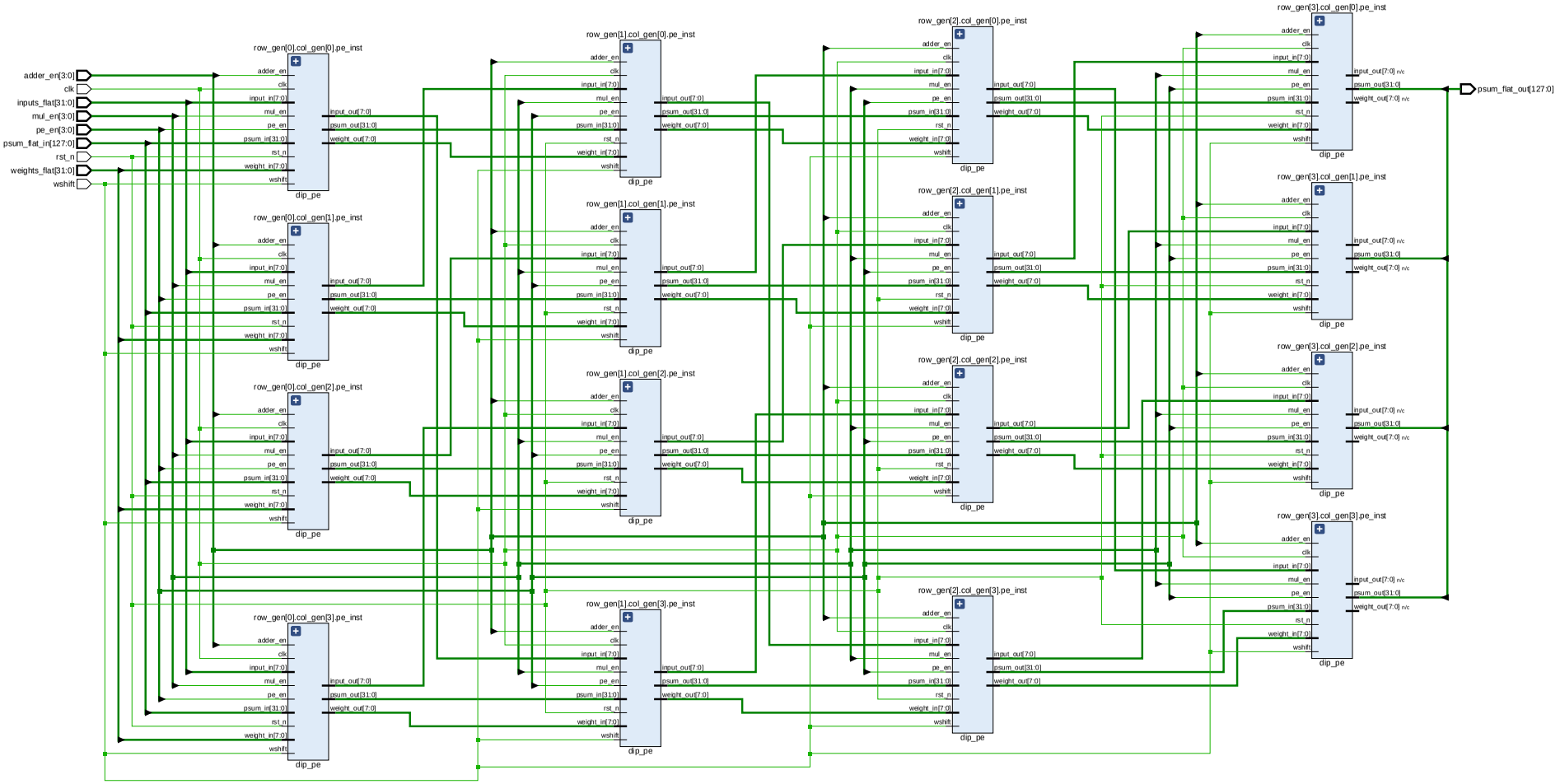

adder_en[3:0]
clk
inputs_flat[31:0]
mul_en[3:0]
pe_en[3:0]
psum_flat_in[127:0]
rst_n
weights_flat[31:0]
wshift
psum_flat_out[127:0]
row_gen[0].col_gen[0].pe_inst
row_gen[0].col_gen[1].pe_inst
row_gen[0].col_gen[2].pe_inst
row_gen[0].col_gen[3].pe_inst
row_gen[1].col_gen[0].pe_inst
row_gen[1].col_gen[1].pe_inst
row_gen[1].col_gen[2].pe_inst
row_gen[1].col_gen[3].pe_inst
row_gen[2].col_gen[0].pe_inst
row_gen[2].col_gen[1].pe_inst
row_gen[2].col_gen[2].pe_inst
row_gen[2].col_gen[3].pe_inst
row_gen[3].col_gen[0].pe_inst
row_gen[3].col_gen[1].pe_inst
row_gen[3].col_gen[2].pe_inst
row_gen[3].col_gen[3].pe_inst
dip_pe
adder_en
clk
input_in[7:0]
mul_en
pe_en
psum_in[31:0]
rst_n
weight_in[7:0]
wshift
input_out[7:0]
psum_out[31:0]
weight_out[7:0]
n/c